CSCI 408: Senior Project I

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Team 9

Final Report

Department of Computer Science

Final Report

Introduction

Qadam is our project based on smart campus navigation with real-time positioning and adaptive accessibility routing, ensuring connection between all university online processes to one understandable platform for students, staff, and visitors who want to navigate in large university campuses. Our team researched the issues that first-year students face and discovered that many of them relate to physical and service navigation caused by a lack of a unified system.

Looking at this, our main objective is to develop an application that focuses on navigation, both indoor and outdoor, with accessibility-aware routes and student onboarding with live updates on campus events, busy spots, and campus business reviews and discounts. Not only that, but the app will have education-related calculators, planners, and cross-departmental updates on services. We aim to make Nu more accessible and convenient to all by introducing one easy-to-use platform.

Background

Indoor Positioning Technologies

Wi-Fi positioning remains one of the widely used indoor positioning methods through wireless network infrastructure. Wi-Fi positioning systems rely mostly on received signal strength fingerprinting and provide a location accuracy of about 1-5 meters (Zafari et al., 2019). Although Wi-Fi positioning has the benefit of widely available infrastructure in university settings, the positioning can be less accurate owing to the variations caused by environmental factors such as the movement of people and the opening/closing of the doors, as well as electronic interference (Kunhoth et al., 2020). The Bluetooth LE beacon positioning technique has attracted many researchers owing to its accuracy of sub-2 meters (Zafari et al., 2019). In research at the FHWS University, results showed that the indoor positioning of users can be accurately achieved through the placement of approximately 3-4 beacons within a single room at a positioning accuracy of about 2 meters (Schäffner & Wilkening, 2022). A hybrid positioning system is far better than positioning involving only one technique. A study has shown that seamless transition from indoor to outdoor positioning can be achieved through GPS positioning of the device when it’s outdoors and Wi-Fi location estimation when indoors using beacons (Alakhras et al., 2015). PDR positioning involves the identification of the device's location through step detection and direction measurement without the aid of an internet connection at a positioning accuracy of about 3-5 meters (Zafari et al., 2019). This type of positioning system can be useful when no internet coverage exists.

Campus Navigation Systems

Some universities already employ indoor navigation solutions and provide significant information regarding the approach and requirements of users. The AlmaWhere system at the University of Bologna reflects a full-scale approach to supporting university campus navigation in particular and focuses especially on accessibility for the disabled (Prandi et al., 2021). The

AlmaWhere research demonstrated the necessity of unique requirements of various groups of disabled users to adapt the software interface accordingly. The University of Applied Sciences Würzburg-Schweinfurt (FHWS) in Germany used ArcGIS Indoors software to create the university campus navigation system and carried out profound research about the best practices of BLE beacon placement (Schäffner & Wilkening, 2022). Both researchers highlighted the necessity of the implementation of routes of emergency evacuation and wheelchair ways in the routing tasks. NavCog has been introduced to aid visually impaired individuals and provides strong navigation support concerning applications targeted at accessibility (Kim et al., 2018). The research involved six visually impaired users and took place within the university campus environment and demonstrating the effectiveness of NavCog in guiding the users to the target points while missing the turn when users did not hear the navigation instruction (Kim et al., 2018).

The shortcoming of the existing systems can be explained as follows: first of all, the existing systems only target a singular purpose – general routing functionality and accessibility functionality combined, meaning there's no provision of a comprehensive platform that comprises various campus services together in one. Additionally, there's limited development of real-time information about dynamic road blocks that might comprise events, road constructions, and crowds. There's no adaptation of the comprehensive campus systems. Lastly, the development of the system doesn't consider the factors of protecting users' privacy first.

Artificial Intelligence and Machine Learning Applications

Natural Language Processing (NLP) algorithms can be applied to sentiment analysis of user reviews about campus facilities, such as cafeterias, study rooms, classrooms, and campus services. The state-of-the-art transformer models of the BERT family can reach the level of accuracy above 85% in sentiment classification of educational text when fine-tuned properly (Devlin et al., 2019). Sentiment classification algorithms classify reviews as positive, negative, or neutral and extract specific points mentioned in the review (e.g., quality of food, level of noise, availability of seating). Anti-spam filters handling review duplication and bot reviews must be used to ensure review quality and authenticity (Jindal & Liu, 2008). Machine learning algorithms used in crowd prediction models can predict the number of users in campus spaces at a particular point in time based on usage patterns and various campus schedules (class schedules and campus event schedules). Algorithms used can recognize patterns of campus usage at various campus times of the day. For example, algorithms can recognize patterns of campus crowding at class transition times and times of reduced campus usage during the examination periods (Kim et al., 2020). These predictions can provide results proactive enough to provide users with alternative routes, avoiding crowded campus spaces. However, campus usage patterns must be large enough to provide accurate predictions of campus crowding. User behavior similarity algorithms applied in recommendation algorithms can identify users who behave similarly and recommend facilities liked by users who are behaviorally similar to them. Content-based filters can match users' preferences to facility attributes in campus facilities (Adomavicius & Tuzhilin, 2005). In many cases, algorithms combining similarity and content

recommendation algorithms can produce the best results. Privacy-Preserving recommendation algorithms applied can preserve users' facility preferences without revealing users' facility preferences (Yang et al., 2019).

Multi-criteria routing algorithms optimize routes while trading off various factors: shortest distance, accessibility, avoiding crowds, and safety. Standard shortest routes algorithms, like Dijkstra's algorithm and A* algorithm, can be modified to support multiple weighted objectives. This allows the computation of Pareto-optimal paths to optimize various trade-offs (Tarapata, 2007).

Research Gaps and Project Positioning

Although existing research has achieved great progress in indoor localization, campus navigation, AR visualization support, and AI-enhanced services, there are several crucial research gaps that invite the development of the current research. Firstly, although existing research has achieved great progress in the above four development topics, the development of existing research has almost exclusively concentrated on single-purpose campus navigation support solutions. Existing research has almost exclusively concentrated on developing solutions regarding campus navigation support and accessibility support without fully combining the two topics. This results in users needing to access different applications in order to benefit from different campus services. In addition to this disadvantage of requiring users to access different applications in order to benefit from campus services, existing campus navigation support research has almost exclusively concentrated on developing solutions without being concerned about accessibility support when guiding users along routes. This has resulted in users requiring additional support concerning accessibility support when accessing campus navigation support solutions. Even though research has already shown the necessity of accessibility support in campus navigation support solutions through research carried out by Prandi et al. (2021), accessibility support concerning this type of campus navigation support solution has been insufficiently developed. Additionally, existing campus navigation support research has almost exclusively concentrated on developing campus navigation support solutions without being concerned about the real-time integration of campus conditions regarding event activities, campus constructions, facility closures, and campus densities. Existing research has insufficiently explored the development of campus navigation support solutions concerning the real-time integration of campus event activities and campus constructions. Finally, the development of existing research has insufficiently explored campus navigation support solutions regarding the development of location technologies that allow users of campus navigation support solutions to benefit from the functionality of location technologies while protecting users' location information from being accessed without users' authorization. Although existing research has already achieved great progress in the development of campus navigation support solutions regarding the development of location technologies, due to rising location information leakage issues and the requirements of location information regulations, including general data protection regulations and the General Data Protection Regulation.

Product Description

System Features

Our mobile application called Qadam is able to gather information into one place: navigation, campus utilities, functionality for people who need accessibility features, and real-time campus intelligence. The system has five main parts that contribute to system features.

1. **Navigation Module**
   1.1 Hybrid Outdoor-Indoor Positioning
   - GPS-based outdoor positioning with an accuracy of +- 3 meters
   - Indoor Positioning via Wifi and Bluetooth services

   1.2 Multi-criteria Routing
   - Optimization of the route by shortest distance, least crowded paths, and based on other contributing factors
   - Changing the route in real-time through identifying temporary closures of some facilities, events, and the density of the crowd

   1.3 AR Navigation
   - AR-based indoor/outdoor guidance using ARCore/ARKit at a stable 60 FPS.
   - Landmark identification for smoother orientation indoors
   - Navigation assisted by visual cues for real-world objects
2. **Accessibility module**
   2.1 Accessibility Routing
   - Identification of elevators, tactile paths, and steeper places
   - Avoidance of certain objects as inaccessible entrances, stairs, and areas under maintenance

   2.2 Inclusive Guidance Tools
   - High-contrast mode, scalable text, and personalized UI
   - Offline emergency routing with clear instructions
3. **Campus Services**
   3.1 Academic tools
   - A GPA calculator that calculates your average grade through the courses registered by the user and the defined grades by them
   - An Academic Planner that is synchronized with the Registrar, showing the courses for the next semester, and a preliminary schedule

   3.2 Campus Utilities
   - Real-time Campus Map with classroom numbers, dining areas, libraries, and other facilities
   - Booking System for study rooms

   3.3 Student Engagement Attributes
   - The list of student discounts provided by the Ministry of External Affairs

- Event board with the upcoming events

4. Social Features

   4.1 Feedback System
   - An AI-powered review system that calculates the overall experience of users
   - Providing feedback through binary selection - like/dislike for one particular place

   4.2 Personalized Recommendations
   - Place recommendations that are generated through the previous experience of the user: the noisiness of study spots and the most visited cafes at the university
   - Filtering by the conditions given by the user at the start of the search

5. System Security
   - Offline-first Design: cached building maps and pre-defined routes
   - Local storage of recent schedules
   - End-to-end encryption of sensitive data

Requirements and Functionalities

The system requirements for Qadam are derived from NU students' experiences as the target audience and their needs.

Functional Requirements

1. Positioning and Localization

- GPS positioning with approx. ±3 meters accuracy outdoors.
- GPS positioning based on Wi-Fi triangulation and Bluetooth beacons with approx. 1–2 meters indoors.
- GPS automatically transitions between indoor and outdoor without user input.

2. Routing and Real-Time Adaptation

- Routing that addresses:
  - Shortest path to the destination,
  - Accessible paths to the destination - ramps, elevators, construction,
  - The least crowded path.
- Live adjustments triggered by:
  - construction,
  - event occurrences,
  - crowd density affected by path adjustments,
  - deviations from the suggested path.
- Saved cached maps and pre-computed evacuation routes for offline usage.

3. AR-Based Navigation

- Directional navigation with AR-based arrows, markers, and landmark recognition.
- Automatic use of 2D navigation for devices without AR support.

4. Reviews and Sentiment-Based Recommendations

- Page for review database for campus businesses and services.
- NLP to analyze sentiment in user reviews.

- Mechanisms to predict and prevent spamming occurrences and weighted reputation scoring for online services.
- Recommendations based on personal inputs and filters for study locations, dining options, and campus events.

5. The System of Discount and Rewards

- A check for the eligibility of each student for discounts
- Loyalty points are awarded for reviews and app engagement.
- The distribution of loyalty points for each review and app-user interaction
- When appropriate, integration with external vendor APIs

6. Academic Tools

- GPA calculator
- An academic planner, which checks the credits earned and the courses that a student should take
- Academic schedule syncing with the calendar

7. Authentication and Security

- JWT token-based sessions and secure OAuth 2.0 authentication
- End-to-end encryption for sensitive data
- End-to-end encryption of sensitive data
- Compliance with relevant data protection and privacy standards
- Conformity with data protection and privacy

8. Multi-Platform Software

- Mobile Application supporting iOS and Android operating systems
- UI for a successful user experience, which is guided by native design principles
- Accessibility support through text size editing, high-contrast mode, and other features

Non-Functional Requirements

1. Performance

- The generation of the route will be at most 2 seconds
- The mobile application will be able to support at least 1000 users

2. Reliability and Availability

- Aim for a system uptime of over 99%
- Recovery mechanisms that are automated in case of failures

3. Scalability

- Cloud-based microservices architecture supporting horizontal scaling.
- Modular services for routing, reviews, authentication, and analytics.

4. Security

- Regular security audits and logging.
- Differential privacy for aggregated analytics.

5. Usability

- WCAG 2.1 AA accessibility standard compliance.

- Intuitive interface for first-time users and visitors.

6. Domain Requirements

- Accurate multi-floor maps with accessibility metadata (ramps, elevators, restricted rooms).
- Integration with the campus course system and event schedule feed.
- Support for accessible evacuation routing during emergencies.
- Compliance with campus data handling policies for academic and personal data.

Design

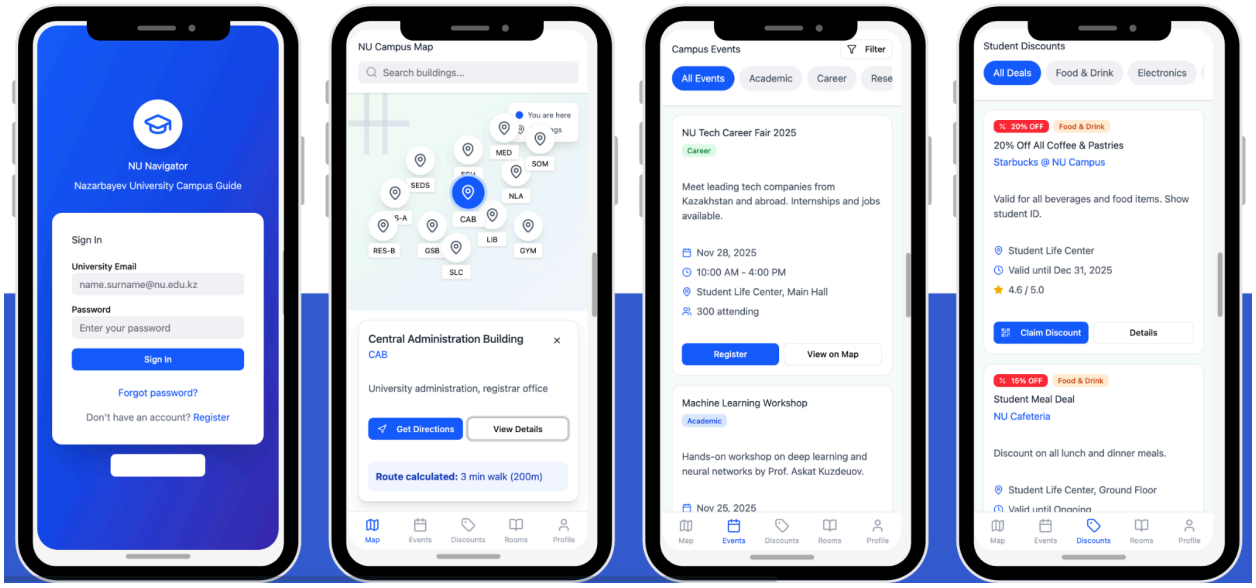

Figure 1. Preliminary design of the mobile application

The design of our Qadam application was defined through the combination of white and blue colors, and minimalistic features for a successful user experience. Figure 1 shows the login page for entering the username and password and sign-in option, a map that shows the location of student blocks and dining places, events page (where upcoming events will be featured), and student discounts page showing the list of services provided and details of each discount.

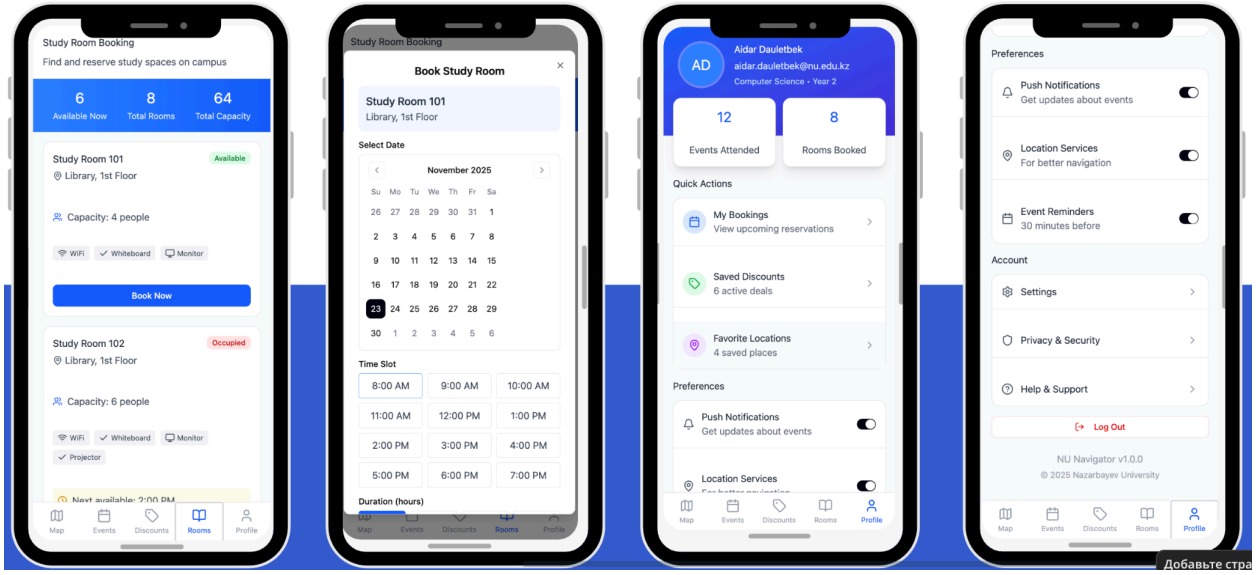

Figure 2. Preliminary design of the inner functionality of the mobile application

Figure 2 displays the social features and academic utilities as study room booking and profile section. In the Profile section, the user may set the location services, notifications, and event reminders. Also, in settings, the user is able to edit the information about themselves.

Progress this Semester

Methodology

For the development of the prototype for this semester, our team employed an Agile approach with weekly meetings and task tracking in order to keep the progress ongoing in Notion. Each member focused on their specific part of the system and regularly met up to review the progress or make adjustments.

The mobile application itself was built on Flutter and tested on both IOS and Android simulators. We experimented with Docker containers to handle real-time data and authentication, and we began setting up PostgreSQL for storing structured information.

For version control, all code is managed through GitHub, and we plan to integrate Firebase later in the project for deployment testing and additional backend functionality. Design mockups are created in Figma, and we use JIRA-style Kanban boards in Notion to manage tasks and deadlines.

Implemented Features

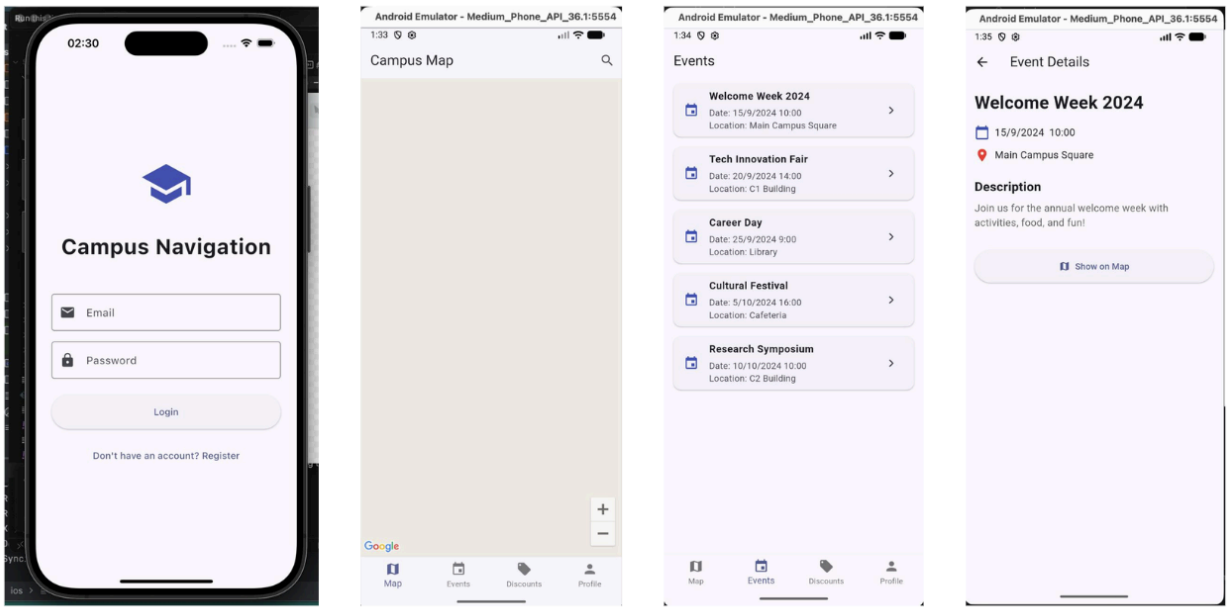

Figure 3. The mobile prototype

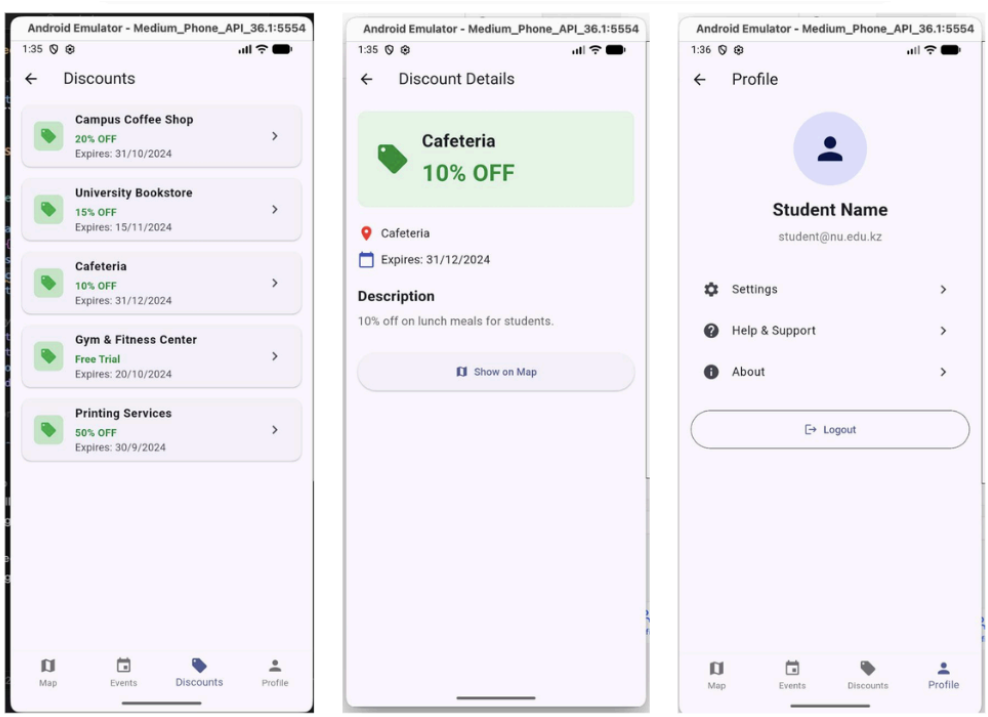

Figure 4. The rest of the panels of the mobile prototype

During this semester, a demo version of our application was created, which stores some information and provides feedback. Firstly, we implemented a login section, which contains registration and logging in with an already existing account. Secondly, we prepared a cascade version of the map that will be further developed. Thirdly, there is a list of cafes, beauty salons,

and other services that provide discounts for NU students. Also, we defined the profile section, where users can edit personal information and set the preferences for notifications and the theme of the mobile app.

Besides the demo version of the mobile application, we assembled backend foundations. A backend skeleton is able to provide basic services for authentication, map within routing, and a discount system. Most importantly, the design of the initial database schema is outlined: tables for users, their profiles, locations of facilities, routes from one place to another, discounts, and reviews given by the users. Finally, API specifications were also defined to communicate within the necessary frames.

Overall, the Fall semester was productive in terms of the delivery of a functional prototype with authentication, a profile page with settings, a panel for discounts, preliminary map design, with a backbone of backend structure that will smoothly pass the data.

Encountered problems

1. **Indoor positioning complexity**
    - Indoor positioning algorithms turned out to be more complex than expected.
    - We did not have access to the needed infrastructure for Bluetooth beacons and Wi-Fi positioning.
    - **We solved this by** relying only on GPS positioning for routing.
2. **Backend challenges**
    - Implementing Django backend, database schemas, and 40 API endpoints took more time and effort than expected.
    - Not deploying our backend slowed the API integration process in the mobile app.
    - **We solved it by:**
        i. Building the backend skeleton to ensure basic capabilities.
        ii. Writing out basic API components and communicating with the mobile app team.
3. **Mobile app challenges**
    - Setting up Flutter for both IOS and Android required additional configurations.
    - Features that depended on backend API endpoints slowed the process of development.
    - **We solved it by:**
        i. Researching Flutter configs for different mobile applications
        ii. Focusing on core mobile app functionalities while waiting for API endpoints.
        iii. Using placeholder data in features that require APIs.

Plans for next semester

Next semester, our team will shift from basics to the actual implementation of our system. The main focus is going to be on transforming the prototype into a functional platform.

On the backend side, we will build the routing engine, integrate hybrid positioning methods, and improve database performance. This will make everything faster and more reliable. We also plan to connect the system to university services like course schedules, calendars, and authentication. Machine learning features will become more complex and will be integrated into backend analytics.

On the other end, the mobile team will finish the development of tools like the GPA calculator, planner, and discount system, and add AR navigation. We will focus on making the app accessible on both iOS and Android.

The next semester will use more testing to ensure the system is working smoothly with a variety of different types of users and situations, including those using features for accessibility.

All in all, this semester laid a very solid grounding for our future work.

Ethical & Legal Considerations

Identification of Ethical Issues

The major ethical issue is the collection of location data and constant monitoring of users, as this information discloses very personal aspects of people's lives, such as their daily activities, habits, and even where they meet people (Zafari et al., 2019). In a university setting, such tracking may lead to the revealing of delicate information about students' presence in classes, places of studying, and their overall movement on campus, thus creating a dilemma between monitoring under the pretext of navigation and surveillance. Accessibility for all people digitally is yet another important ethical aspect, as university students with disabilities would be effectively excluded if proper accessibility features were not provided, which would be against the principles of educational justice (Prandi et al., 2021). Moreover, it is hard to get truly informed consent in the case of mobile applications, as users usually give permissions without realizing the consequences of their actions, which is made worse by the consent fatigue experienced from having to repeatedly respond to permission requests from different applications.

Legal Compliance

The project is faced with a difficult situation that has a regulatory framework with different legal requirements that must all be complied with at once. The General Data Protection Regulation (GDPR) (EU) 2016 regards location data as a personal data category that is heavily protected and thus requires user consent, which, according to the GDPR, has to be very clear, requiring that consent be freely given, specific, informed, and unambiguous through clear affirmative action. GDPR Article 17 gives users the right to be forgotten, which implies that all personally identifiable location data has to be deleted upon request, whereas Article 5 requires data minimization—collecting only the minimum location data necessary for the stated purpose.

Informed Judgments and Trade-offs

The team responsible for development took many important decisions that were weighing heavily on the scales of ethical principles, legal requirements, user experience, and technical feasibility. The most basic decision was to go for a privacy-by-design architecture that included privacy considerations in the system from the very beginning through default-off location

tracking that required the user to actively enable positioning services. Personalization and privacy were seen as a significant trade-off, which was tackled by implementing a tiered permission system where users selecting the anonymous mode method would be given basic navigation functionality without any data collection, while those opting for personalized features would have to give their consent in a specific way for data use. The AR navigation part brought up some compatibility issues, as ARCore and ARKit technologies can still be used on the latest devices only, which made it necessary to come up with 2D overlay fallback systems that guarantee universal access while still giving priority to the inclusive access over the premium features (Romli et al., 2024).

Sustainability and Social Responsibility

The Smart Campus Navigation project is grounded in social responsibility principles, which not only help individual users but also have a positive impact on society by providing the whole range of accessibility features, which guarantee that the visually impaired, the deaf, the physically impaired, and the intellectually challenged students to move around the campus by themselves. The accessibility-aware routing algorithm, which marks out wheelchair-accessible paths, tells users the location of elevators and provides maps of all the routes with no barriers at all, is a great help to mobility-impaired people and makes the campus navigation easier for them. Research indicates that such assistive technologies considerably cut down the navigation time and boost the independence of users with disabilities physiologically (Prandi et al., 2021).

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